



Visualization of transport pathways for organic compounds in undisturbed soil monoliths

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ABSTRACT

Small amounts of organic compounds are often found at much larger depths than expected from their organic carbon normalized distribution coefficient (K_{oc}). In this study the food dye Brilliant Blue (BB) was used to visualize the (preferential) transport pathways in undisturbed soil monoliths taken from an Orthic Luvisol and from a Eutric Cambisol. After spiking the monoliths with 1.0 MBq of either ^{14}C -labeled benazolin (leacher) or ^{14}C -labeled benzo[a]pyrene (BaP, nonleacher), they were intermittently irrigated under free-draining conditions twice weekly with 4 mm of rain water for 17 weeks. BB (4 g L^{-1}) was added to the rain water during the last 4 weeks. After irrigation, the monoliths were sliced and photographs were taken from horizontal cross-sections. Soil was dried and incinerated to measure total ^{14}C -activity. The mean travel depth for benazolin (6–7 cm) was larger than for BaP (2 cm), which is in line with their K_{oc} values, although BaP was much more mobile than expected from the K_{oc} value. BB patterns, arbitrarily classified into five intensity classes, showed that large parts of the monolith were bypassed below the completely stained upper two to five centimeter. Furthermore, BB patterns indicated the locations of strongly sorbing compounds in the deeper soil layers and are therefore a helpful tool to selectively sample the soil for these compounds. A continuously stained root channel below the 20-cm depth contained substantial amounts of BaP, indicating that soil structure cannot be neglected to assess the fate of nonleachers in undisturbed soils.

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1. Introduction

Small amounts of xenobiotics are often found at much larger depths than expected from their chemical properties (Flury, 1996). For nonpolar, non-ionisable compounds, strong positive correlations have been observed between sorption strength, expressed as the distribution coefficient, K_d , which characterizes the partitioning of a solute between the liquid and the solid phase, and total soil organic carbon (e.g. Mermoud et al., 2008). This led to the development of the widely-used approach of the organic carbon normalized distribution coefficient, K_{oc} , where the distribution coefficient, K_d , was normalized by the total organic carbon content, f_{oc} , thus decreasing the variability of the distribution coefficients between different sites. Implicit assumptions for the application of this bulk soil approach to assess transport velocities of organic contaminants are local sorption equilibrium (Valocchi, 1985) and uniform water flow.

However, several objections can be raised regarding the normalization of the distribution coefficient by the total organic carbon content. First, not only the quantity of the soil organic matter, but also its nature,

i.e. the composition of organic matter regarding the functional groups, affects sorption (Grathwohl, 1990). For example, Hawthorne et al. (2006) still found a variation in sediment-measured K_{oc} values of up to three orders of magnitudes for several polycyclic aromatic hydrocarbons, including benzo[a]pyrene. Ahmad et al. (2001) considerably reduced the variation in the normalized distribution coefficient by using aromaticity instead of f_{oc} . Second, the strong relationship between the distribution coefficient and the soil organic matter weakens under flow conditions (Elabd et al., 1986; Lennartz, 1999). This is especially the case when structure-dependent fast transport pathways, such as earthworm burrows, roots, cracks, and fissures, are active (Flury, 1996; Kasteel et al., 2010), thereby violating at least the assumption of a uniform flow field. The activation of preferential flow pathways is affected by climatic factors (rainfall intensity), soil management practice (application date, tillage), and history of the water status in the soil (Gerke, 2006; Jarvis, 2007). Local water fluxes in these preferential flow pathways are much higher than those assumed for a uniform flow field, which leads to high local transport velocities. High transport velocities may lead to chemical non-equilibrium and less sorption compared to the bulk soil. Thus, apart from bypassing a part of the bulk soil, the non-compliance with the assumptions of a local equilibrium and a uniform flow field results in a much faster transport of solutes through soils than might be expected based on the K_{oc} values. Third, the normalized distribution coefficient K_{oc} is often inaccurately applied to organic compound where soil organic matter may not be the key sorbate. For

Abbreviations: MRZ, Merzenhausen; SCH, Scheyern; BaP, benzo[a]pyrene; BB, Brilliant Blue; TLC, thin layer chromatography.

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example, the mineral phase (mainly oxides) is responsible for the strong sorption of glyphosate in soils, not the soil organic matter (Vereecken, 2005). Fourth, chemical properties of the soil particles and organic matter in preferential flow paths are different from the bulk soil (e.g., Leue et al., 2010). At last, there is a discrepancy between laboratory-derived K_{oc} values and field-derived K_{oc} values due to differences in contact times (Vereecken et al., 2011).

The food dye Brilliant Blue FCF (Colour Index 42090) is a commonly used tracer for the visualization of the transport pathways of water in the vadose zone (Flury et al., 1994). It is often continuously applied at the soil's surface with the invading fluid using rather high input concentration in the order of gram per liter to get a visible contrast with the non-stained soil. In this application mode, Brilliant Blue (BB) is considered to be a rather mobile compound, in contrast to a pulse application of similar input concentration followed by a subsequent dilution of the concentration by a tracer-free solution (Kasteel et al., 2002). Now, Brilliant Blue can be considered as a moderately to strongly sorbing compound. Digitized images of dye patterns are evaluated in either a qualitative, in a semi-quantitative, or in a quantitative way to interpret the flow and transport behavior in soils. Where initially the analysis of the dye patterns was restricted to binary images, i.e., a pixel is either stained or unstained (Flury et al., 1994; Petersen et al., 1997), it evolved to a semi-quantitative method, where pixels are assigned to dye intensity classes (Cey et al., 2009; Weiler and Flühler, 2004), and to the quantification of dye concentrations in soil (Ewing and Horton, 1999; Forrer et al., 2000; Morris and Mooney, 2004). The information on the dye patterns is then used to interpret the flow behavior.

In a field-plot transport study with ten herbicide and insecticides, Reichenberger et al. (2002) found that, considering mass fractions, the less mobile (apolar) pesticides were more prone to vertical transport by preferential flow than the more mobile (polar) pesticides. They confirmed that the dye BB marked the transport pathways of the pesticides and thus can be used as a tool for an effective pesticide sampling strategy (see also Gjettermann et al., 2011). Also Burkhardt et al. (2008) and Cey et al. (2009) used the staining pattern of BB to guide the selective sampling of the hydrologically active preferential transport pathways for microspheres, which are frequently being used as a substitute for bacteria and colloids and which can act as a carrier for strongly sorbing pesticides.

This study comprises controlled leaching experiments with benzo[a]pyrene (BaP) and the herbicide benazolin in combination with BB in undisturbed soil monoliths. BaP is a nonpolar organic compound with a high sorption affinity to the soil's organic phase and is, as a higher molecular weight polycyclic aromatic hydrocarbon, quite resistant to microbial decay. Its pK_{oc} value is in the order of 5 for humic acids extracted from four typical Chinese soils (Zhang et al., 2009), with pK_{oc} values varying from 4.8 to 6.4 for soil from a plow layer (Kögel-Knabner et al., 2009), and with pK_{oc} values ranging from 5.5 to 8.3 for different sediments with a median pK_{oc} value of 6.65 (Hawthorne et al., 2006). The post-emergence acidic herbicide benazolin (4-chloro-2-oxo-benzothiazolin-3-yl acetic acid) is the first transformation product of benazolin-ethyl. The laboratory-determined half-life value, DT_{50} , of this transformation is less than one day (Leake, 1989). Benazolin itself is also not stable and is further degraded with laboratory-determined half-life of two–four weeks into thiazolin (4-chloro-benzothiazolin-2-one), which has similar transport properties as benazolin (Leake, 1989). In transport studies, benazolin-ethyl was less mobile compared to benazolin (Leake, 1989), the latter having a pK_{oc} value in the order of 1.34 (Goudarzi et al., 2009). The pesticide property database PPDB (2011) reflects the sorption and degradation behavior of benazolin-ethyl (pK_{oc} = 3.12 and DT_{50} = 1.5 d) and benazolin (pK_{oc} = 1.56 and DT_{50} = 21 d).

Jury and Gruber (1989) classified pesticides based on their different environmental fate properties, i.e., sorption and degradation, into low- and high-risk leaching classes to assess their leaching potential below the microbial active topsoil. Following this classification, benazolin is a

potential leacher, whereas BaP is a virtually immobile nonleacher. The objective of this study is to verify the assessment of the travel distance of the leacher benazolin and the nonleacher BaP in undisturbed soil monoliths of two distinct top soils based on the approach of organic carbon normalized distribution coefficient by measuring bulk soil and local concentrations. For the determination of local concentrations, the dye tracer BB is used to visualize the hydrologically active transport pathways which then can be used to guide a selective sampling of the pesticides. Benazolin and BaP are both ^{14}C -labeled, which allows the determination of bulk soil ^{14}C -activity by combustion without additional extraction steps.

2. Material and methods

2.1. Experimental setup

The leaching experiments with the dye tracer BB were conducted in the framework of a larger study on the effect of maize residues on the fate of benazolin in undisturbed soil monoliths (Berns et al., 2007; Schnitzler et al., 2007). We refer to these two studies for more detailed information on the experimental setup. For this study, in short, two undisturbed soil monoliths (diameter is 20 cm; length is 40 cm) were taken from a silty loam (Dystric Luvisol) near Merzenhausen (MRZ), North Rhine–Westphalia, and one from a loam (Dystric Cambisol) near Scheyern (SCH), Bavaria, both in Germany. The soils were classified according to the World Reference Base for Soil Resources (FAO, 2006). Both sites were located on arable land used for agricultural production. The plow layer of the silty loam (0–35 cm depth) consisted of 4% sand, 79% silt and 17% clay and that of the loam (0–25 cm depth) of 42% sand, 36% silt and 22% clay measured by the pipette method after oxidation of the organic matter with H_2O_2 (Berns et al., 2007). The silty loam had an organic carbon content of 1.0% in the plow layer compared to 1.4% in the loam. A plow pan was existent in all monoliths, but it was considered to be of minor importance for the transport of the mean travel distance of organic compounds, since it was either of a limited extent, starting close to the artificial zero-potential lower boundary (Merzenhausen) or the applied tracer was expected to show a very strong interaction with the solid phase (Scheyern). The occurrence of organic compounds in the drainage water was expected to happen only through transport in macropore structures, consisting of biopores, inter-aggregate structures and cracks, which were more abundant in the Scheyern soil.

Steel cylinders with a sharpened lower edge were pressed from the soil surface into the soils by a hydraulic drilling system. The monoliths were operated in the dark at a constant temperature of 12 ± 1 °C. In line with the maize-amended treatments of the larger study, the top 5-cm of the monolith was loosened by a spoon, after removing possible plant residues from the soil surface. This homogenization also reduced potential wall bypass flow in the upper part of the soil monoliths. Before operation, the monoliths were closed to prevent loss of water by evaporation.

All monoliths were irrigated twice weekly with 4-mm of rain water for 17 weeks. No ponding was observed at the subsequent water application. Drainage was allowed by gravity. During the first irrigation event starting from field-moist conditions close to field capacity, the monoliths from MRZ were spiked with 1.0 MBq ^{14}C -labeled benazolin and the monolith from SCH with 1.0 MBq ^{14}C -labeled BaP using an application solution. A soda lime trap was only mounted on the benazolin monoliths to detect mineralization of the applied ^{14}C -benazolin. The trap was removed and analyzed during the bi-weekly irrigation events (Schnitzler et al., 2007).

Irrigation resulted in a cumulative amount of drainage water of 13.9 cm for the two MRZ soils compared to 13.3 cm for the SCH soil. Cumulative drainage expressed as a fraction of the total water input was 1.02 for the MRZ soil compared to 0.98 for the SCH soil (see also Fig. 4c). These values demonstrate that evaporation losses were

negligible and that the initial moisture status was close to field capacity. For the last 4 weeks, the rain water – which was applied bi-weekly – was spiked with 4 g L^{-1} of the dye tracer BB. This application mode for BB can be regarded as a step application, since no dilution with dye-free irrigation water occurred at the soil surface. The weekly collected drainage water was analyzed for ^{14}C -activity using a liquid scintillation counter (Schnitzler et al., 2007). The BB concentration was measured with a UV photo-spectrometer (Uvikon 860) at a wavelength of 630 nm.

Upon completion of the tracer application, horizontal cross-sections were prepared, at one to five centimeters depth intervals. The surfaces of these cross-sections were photographed with a digital camera mounted on a tripod. The distance between the camera and the surface of the monolith was kept constant. Two flash units were installed, both having an angle of 45° to the soil surface. Photographs of the soil were obtained in color, together with a Kodak color and gray scale, a gray paperboard, and a ruler in the image (see Fig. 1). The cross-sections were selectively sampled by scraping approximately 1.5 g soil from stained and non-stained regions with a knife. These sampling locations were marked with a pin and an additional picture was taken. For each monolith we photographed and sampled 10 to 16 horizontal cross-sections.

The soil slices with known volume were weighted before and after oven-drying at 105°C to determine the volumetric water content, θ , and the dry bulk density, ρ_b . After oven-drying, each slice was homogenized, ground, and the ^{14}C -activity was determined by combustion using an oxidizer (Schnitzler et al., 2007). For each soil slice, ^{14}C -activity per dry mass of soil was converted to ^{14}C -activity per volume of dry soil using its measured dry bulk density. Note that liquid scintillation counting and the combustion method only provide a measure of ^{14}C -equivalents, which consists of the parent compound plus their metabolites.

2.2. Moment analysis

The transport behavior of pesticides and BB was characterized by moment analysis. This is a purely statistical technique for summarizing solute transport without making any assumptions on a specific transport model. Spatial (soil concentration) and temporal (drainage water concentration) moments were numerically evaluated using the trapezoidal rule (Das et al., 2005).

2.3. Image analysis

The pixel number of the original images, as shown in Fig. 1 (3007×1959), was reduced by a factor of four, after smoothing using a median filter with a side length of two pixels. The final image had a pixel size of $0.58 \times 0.58 \text{ mm}^2$. The static photographic setup ensured that images were not geometrically distorted.

Background subtraction based on the value V of a gray area is an effective method to reduce the effect of inhomogeneous illumination (Forrer et al., 2000). We therefore sampled the RGB values on each of the four corners of the gray paperboard and transformed it to the HSV color space. The analysis of the mean value V revealed that, in general, it hardly varied within each image and was nearly constant between the images of the same monolith. Thus, no corrections were performed for inhomogeneous illumination.

The goal of the image analysis procedure is to classify the dye patterns in intensity classes, which are in a non-linear way related to concentrations. Based on Ewing and Horton (1999), who found the closest relationships between the H and V values and the logarithm of the dye concentration, Weiler and Flüher (2004) defined a transformation variable, T ,

$$T = \frac{H}{360\pi}(1-V) \quad (1)$$

where H (rad) is the hue and V (–) is the value which ranges between 0 and 1. All the images were transformed according to Eq. (1), reducing the three-dimensional RGB color space to one dimension.

We used conditional dilation (Serra, 1988) to classify images in stained and non-stained areas. In this method, a lower and upper limit is used for thresholding and the region in between is considered as a fuzzy region. To define the lower threshold limit, we sampled T -values in circular areas with a radius of three pixels in the non-stained areas. The average of these values was used as the lower limit and the upper limit was arbitrarily taken 1.5 times as large. All pixels with a T value lower than the lower limit or with a T value lower than the upper limit and with at least one direct neighbor pixel with a T value lower than the lower limit were assigned to be non-stained. This algorithm is repeated iteratively until no pixel is to be changed. Isolated stained pixels, which have no stained neighbors, are then removed. Finally, the stained areas were arbitrarily

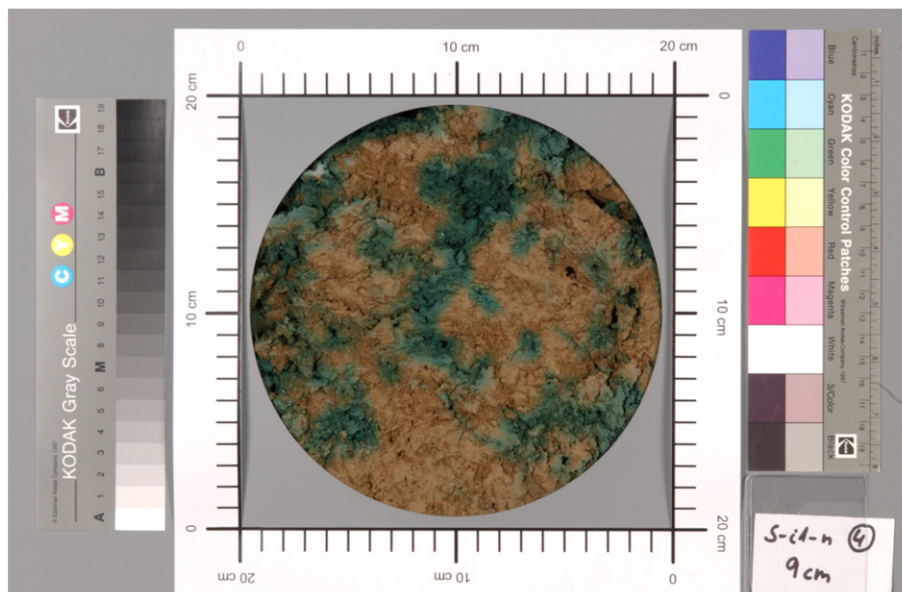


Fig. 1. Example of an original image for the 9-cm depth of the Scheyern soil monolith.

subdivided into four intensity classes related to increasing dye concentrations.

3. Results and discussion

The setup of the experiment with the application of the herbicides in different soil types does not allow to separate out the influence of differences in soil properties from that of differences in the herbicide's chemical properties. The soil monolith studies provide illustrative examples on the interaction of a herbicide with the soil under the prevailing water flow conditions.

3.1. Total ^{14}C -activity profiles

The depth profiles of dry bulk density, ρ_b , and water content, θ , at the end of irrigation are shown in Fig. 2 for the three soil monoliths. Although the upper 5 cm of soil in the monoliths was loosened, bulk density was markedly higher in the upper slice in both MRZ monoliths compared to the SCH soil. The structural stability of the MRZ soil (silty loam) was not large enough to retain the loose structure during irrigation. Dry bulk density was on average larger in the MRZ monoliths ($\rho_b \approx 1.6 \text{ g cm}^{-3}$) compared to the SCH monolith ($\rho_b \approx 1.4 \text{ g cm}^{-3}$). The depth profiles of the volumetric water content, θ , closely followed the course of the dry bulk densities, with on average a lower value in the MRZ monoliths ($\theta \approx 0.27 \text{ cm}^3 \text{ cm}^{-3}$) than in the SCH monolith ($\theta \approx 0.30 \text{ cm}^3 \text{ cm}^{-3}$) (Fig. 2c and d).

Mass recoveries in the soil and in the drainage water are expressed as the ratio of zeroth spatial or temporal moment and the total mass applied per unit area. The overall mass recovery, which is the sum of the zeroth spatial moment, the zeroth temporal moment, and

mineralization, was good with values between 90 and 100% for the two compounds (Table 1). Even for the more mobile benazolin (MRZ), most mass (90%) still resided in the soil after 13.9 cm of cumulative net infiltration – which equals equivalent pore volumes of 1.31 for MRZ#1 and 1.24 for MRZ#2, based on the assumption that all water participated in solute transport – with only minor mass losses due to mineralization (5%) and leaching (1–3%). For the nonleacher BaP (SCH), all mass resided in the soil monolith after 13.3 cm of cumulative net infiltration – representing an equivalent pore volume of 1.08 – with virtually no leaching ($2 \times 10^{-3}\%$).

Fig. 3 shows the concentration profiles of the organic compounds in the soil monoliths. In contrast to BaP, whose bulk soil ^{14}C -activity was above the detection limit only in the upper 25 cm of the soil monolith, ^{14}C -benazolin was detected in all soil slices. BaP showed a higher maximum ^{14}C -activity and a much faster decline of the ^{14}C -activity with depth compared to benazolin. This is qualitatively congruent with the reported much higher K_{oc} values for BaP, which results in a higher retardation. The replicate monoliths of the MRZ soil showed a similar trend in ^{14}C -benazolin activities from the soil surface down to the 20-cm depth. Below, a sudden drop in ^{14}C -activity occurred in monolith MRZ#2 with subsequent increasing concentrations near the monolith outlet. This might be an indication of an artifact associated with wall flow. In contrast, an exponential decline in ^{14}C -benazolin activity over the entire depth was observed in monolith MRZ#1, which is often interpreted as an indication of preferential flow (Kasteel et al., 2005).

3.2. Mean travel depth

Since most mass of benazolin and BaP was retained in the soil, we further interpreted their spatial moments, i.e., the first normalized depth moment or mean travel depth, μ_z , and the second central depth moment or variance, σ_z^2 , which describes the solute spreading around the mean travel depth (see Table 1). The mean travel depth in the two benazolin monoliths ($\mu_z = 6\text{--}7 \text{ cm}$) was a factor 3 to 4 larger compared to the mean travel depth of BaP ($\mu_z = 1.8 \text{ cm}$). This difference in mean travel depth was remarkably small considering the disparity of more than three orders of magnitude in their K_{oc} values. Notice that the slice thickness (1 cm for benazolin and 0.5 cm for BaP in the upper part of the monoliths) was considerably smaller than the first moment. The twofold larger variance of benazolin in soil monolith MRZ#2 compared to MRZ#1 was remarkable and indicates a more pronounced spreading over the monolith. The spreading of BaP is lower by a factor of 5 to 10 compared to benazolin, which qualitatively reflects the lower spreading potential for substances with larger retardation.

Assuming that water flow is uniform in soil and that the sorbed concentration of a reactive compound is in instantaneous equilibrium with the concentration in the liquid phase, i.e., the local equilibrium assumption (Valocchi, 1985), an estimate of the travel depth, $\hat{\mu}_z$, for both compounds in the monolith can be calculated as follows:

$$\hat{\mu}_z = \frac{I}{\theta R} = \frac{I}{\theta(1 + \rho_b f_{oc} K_{oc} / \theta)} \quad (2)$$

Table 1

Mass recovery in the various compartments, mean travel depth, μ_z , and variance, σ_z^2 , determined by moment analysis for the ^{14}C -labeled compounds benazolin in the Merzenhausen soil (MRZ) and benzo[a]pyrene (BaP) in the Scheyern (SCH) soil.

Soil	Compound	Mass recovery (%)			μ_z (cm)	σ_z^2 (cm ²)
		Soil	Outflow	Mineralization ^a	Total	
MRZ#1	Benazolin	91.9	0.5	6.4	98.8	5.7
MRZ#2	Benazolin	81.8	2.9	4.5	89.2	7.0
SCH	BaP	92.7	0.0	n.a. ^b	92.7	1.8

^a Mineralization was measured as $^{14}\text{CO}_2$ with a soda lime trap on the benazolin monoliths.

^b Not analyzed. Mineralization was assumed to be negligible in the benzo[a]pyrene monolith.

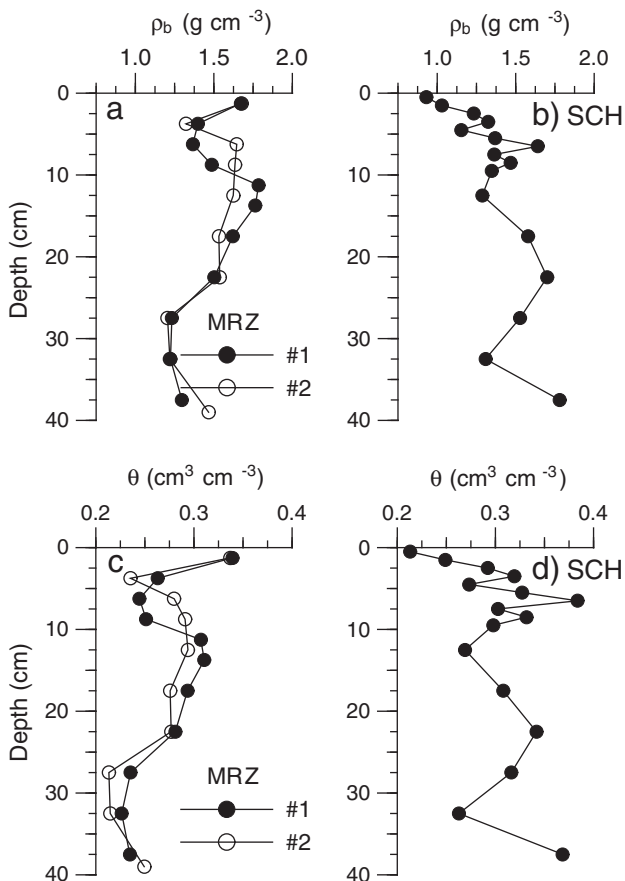


Fig. 2. Depth profiles of dry bulk density, ρ_b , for (a) the Merzenhausen (MRZ) and (b) the Scheyern (SCH) monoliths and of volumetric water content, θ , for (c) the MRZ and (d) the SCH soil monoliths.

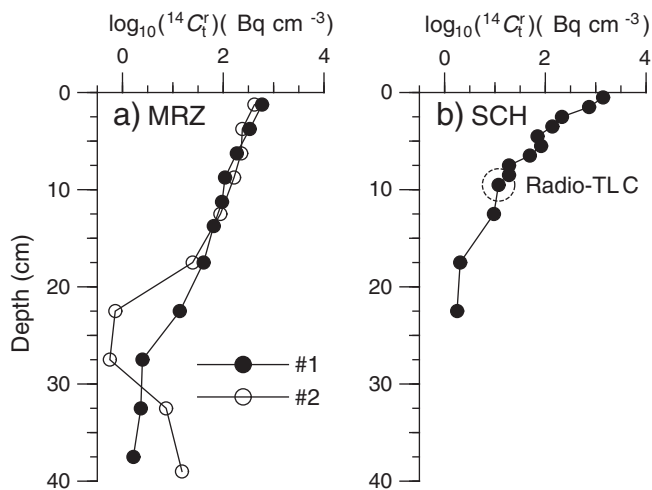


Fig. 3. Depth profiles of the resident concentration, C_r , of (a) benazolin in the Merzenhausen (MRZ) monoliths and of (b) benzo[a]pyrene in the Scheyern (SCH) monolith. The dashed circle indicates the depth where the ^{14}C -benzo[a]pyrene and its metabolites were analyzed by thin layer chromatography (TLC). Notice that the concentration is expressed as ^{14}C -activity per volume of soil.

where I (cm) is the cumulative amount of net drainage and R (–) is retardation factor. Based on reported literature K_{oc} values for BaP, which vary roughly between 10^4 and 10^6 L kg^{-1} , an estimated average organic carbon content of 1.4% (Schroll et al., 2006), and on the experimentally determined average dry bulk density and volumetric water content, the estimated mean travel depth was 6.6×10^{-2} cm for $K_{oc} = 10^4$ L kg^{-1} and 6.6×10^{-4} cm for $K_{oc} = 10^6$ L kg^{-1} . This simple calculation grossly underestimated the measured mean travel depth of 2 cm for BaP.

One possible explanation is that a more mobile metabolite of BaP was responsible for the faster than estimated transport. However, extractions from the bulk soil (soil slice from 9 to 10 cm depth, see Fig. 3b) and subsequent radiolabeled thin layer chromatography measurements showed that between 85 and 95% of the extracted ^{14}C -activity could be assigned to the parent compound BaP, with the remainder of the ^{14}C -activity being divided over one major metabolite, accounting for 1 to 6% to the total ^{14}C -activity, and 3 to 5 peaks of unidentified metabolites with only minor contribution to the ^{14}C -activity. This extraction clarified that mainly the parent compound BaP itself and not a much more mobile metabolite accounted for measured ^{14}C -activity deeper down the soil profile.

More likely explanations for the faster than estimated transport of BaP are a violation of the local equilibrium assumption (chemical non-equilibrium), a violation of the assumption of uniform flow (physical non-equilibrium), or a violation of both. We refer to chemical non-equilibrium, if the sorption processes itself requires time and limits sorption equilibration, thereby accelerating downward transport. We refer to physical non-equilibrium when advection is the limiting factor to reach sorption equilibrium (Gerke, 2006; Jarvis, 2007), i.e., high local flow velocities restrict the contact time between liquid and solid phase. Thus, due to non-uniform flow, part of the applied substance moved downward by advection faster than expected from the mean advection velocity.

The mean travel depth of benazolin was assessed accordingly. Based on the reported literature K_{oc} values of 22 and 36 L kg^{-1} for benazolin, on an organic carbon content of 1.0% (Kasteel et al., 2010), and on the experimentally determined average dry bulk density and volumetric water content (see Eq. (2)), we estimated a mean travel depth of approximately 16 and 22 cm. This threefold larger estimated mean transport distance compared to moment analysis can be explained by the fact that (i) moment analysis slightly underestimated the mean travel depth since it does not account for the ^{14}C detected in the

drainage water; (ii) a less mobile metabolite than benazolin was formed. This was not verified, but thiazolin, which is the successor of benazolin in the degradation chain, has similar transport properties as benazolin (Leake, 1989); (iii) reported literature K_{oc} values are not adequate for this silty loam; and (iv) not the entire bulk soil volume participated to transport, i.e. a physical non-equilibrium, where part of the solute mass was trapped in the bypassed matrix.

3.3. Breakthrough curves

Fig. 4a shows the breakthrough curve of the organic compounds in the soil monoliths. Notice that the mass recovery was at most 3% in the drainage water. Small amounts of ^{14}C -benazolin were quickly transported through the MRZ monoliths, with first detections of ^{14}C -activity in the drainage sample after only 1.5 cm (sampling day 14) of cumulative drainage for MRZ#2 and 2.3 cm (day 21) for MRZ#1. A significant increase in concentration over time in the drainage water occurred earlier and was more pronounced in monolith MRZ#2 compared

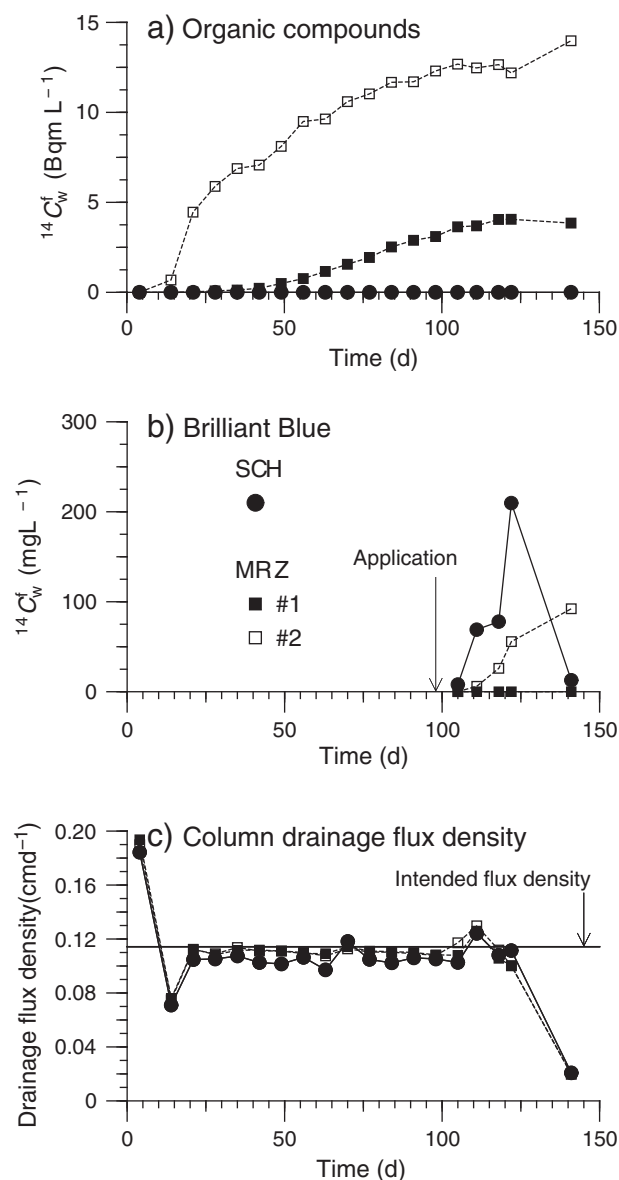


Fig. 4. Breakthrough curves for (a) the organic compounds benzo[a]pyrene in the Scheyern (SCH) soil and benazolin in the Merzenhausen soil (MRZ), and for (b) the dye tracer Brilliant Blue in all monoliths, complemented with (c) the intended irrigation flux density and measured column drainage flux density.

to monolith MRZ#1. This difference may be explained by possible wall flow in monolith MRZ#2, thereby creating a short-cut between the soil surface and the base of the monolith (see discussion on the staining patterns below). In contrast, there were only two positive detections of ^{14}C -activity in the drainage water from SCH monolith (BaP) in the third (day 21) and fourth (day 28) drainage sample (equivalent to a cumulative amount of drainage between 2 and 3 cm), with however only traces of ^{14}C -activity below 0.005 Bq cm^{-3} .

In contrast to the organic compounds with different sorption properties, BB breakthrough was fastest in the SCH monolith and also reached higher concentrations (see Fig. 4b). Apparently, a continuous fast transport pathway existed in the SCH monolith, which was more easily invaded by the mobile BB. This existence of a fast transport pathway is masked by the large sorption potential of BaP. In the MRZ soil, an earlier breakthrough of BB was observed in the drainage water of monolith MRZ#2, with no positive detection of BB in monolith MRZ#1. A net infiltration of 3 cm was apparently not sufficient for BB to reach the soil outlet in monolith MRZ#1. The earlier detection of ^{14}C -benazolin in MRZ#1 might be associated with the higher sensitivity of the ^{14}C -method compared to the photo-spectrometric detection of BB. A substantial increase of ^{14}C -benazolin in the drainage water was not observed in MRZ#1 until 5 cm of cumulative drainage water had passed the monolith.

Fig. 4c shows that drainage flux density was close to the intended irrigation flux density, indicating that evaporation was a minor part of the total water balance. The drainage flux density was fairly constant over time, except for the first two at the last drainage events. The first drainage events show that a quasi hydraulic equilibrium was established after two irrigation events in all monoliths; no water was applied during the last sampling interval, which shows the drainage of the excess of soil water towards field capacity.

3.4. Dye staining patterns

A dye tracer visualizes the soil's transport volume by staining those parts of the bulk soil volume where it is in contact with the solid phase. Fig. 5 shows these staining patterns in horizontal cross-sections at different depths for the two MRZ soil monoliths and Fig. 6 for the SCH monolith. Highest concentrations (black) were observed at the soil surface due to the bi-weekly application of a BB solution.

Dye coverage, which is defined as the ratio of the pixels where a concentration was assigned to by the conditional dilation algorithm and the total number of pixels, was almost 100% in the upper 5 cm for both MRZ monoliths (see Fig. 7a). Full dye coverage coincided with the upper part of the soil monolith that was loosened. Interestingly, dye coverage increased at 15 cm depth in both MRZ monoliths. Furthermore, this increase in dye coverage coincided with an aerial increase of the highest intensity class for BB (Fig. 5). The reason for this increase remained unclear, but it hinted at an impeding layer. At least for MRZ#1, highest dry bulk densities were found at this depth.

From 15 cm downward, dye coverage decreased to virtually zero below 25 cm depth for MRZ#1, which is in agreement with the BB zero concentration in the drainage water collected at the 40-cm depth (Fig. 4b). In contrast, dye coverage remained more or less constant at 25% below 15 cm depth in MRZ#2, which marked a continuous preferred transport pathway with a rather high intensity class adjacent to the cylinder wall. This dye pattern suggests that an artifact, i.e., wall flow, might be responsible for the early arrival and faster increase of ^{14}C -benazolin in the drainage water (see Fig. 4a), and larger spreading in MRZ#2.

Regarding preferential flow, Flüher et al. (1996) distinguished three flow regimes based on phenomenological transport features: (i) distribution flow in the attractor zone where convergence of streamlines are initiated; (ii) preferential flow in the transmission zone with rapid flow and hardly any lateral mass exchange; and (iii) dispersive flow in the dispersion zone with considerable lateral

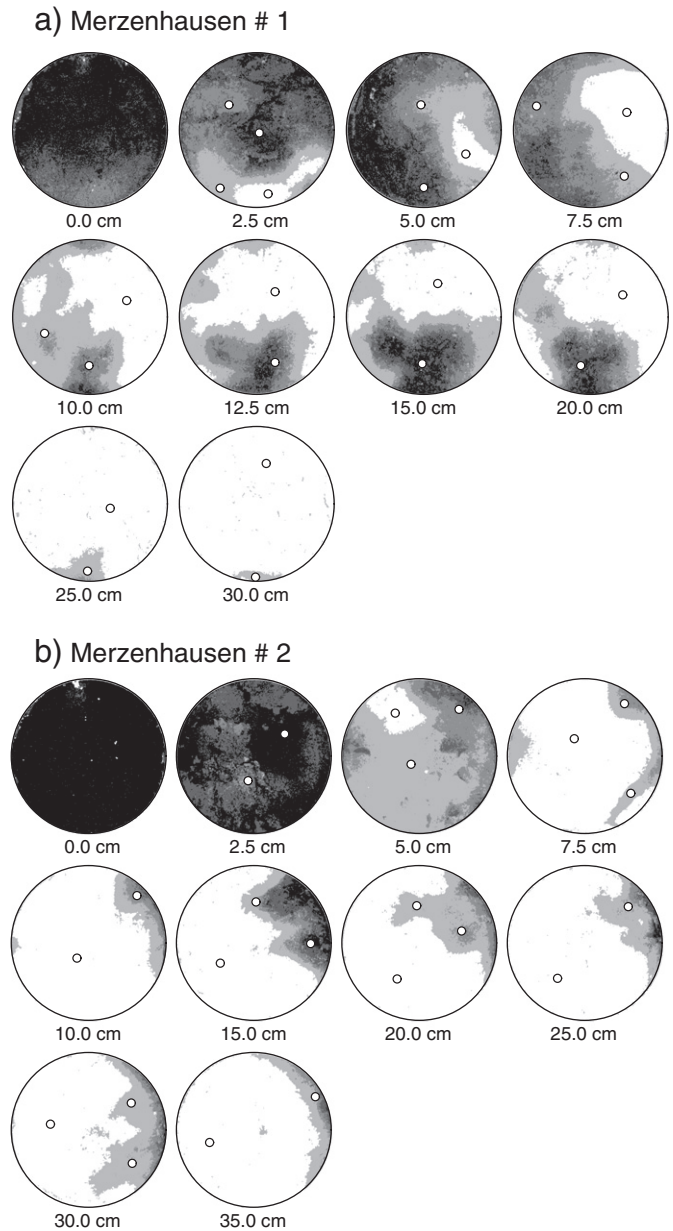


Fig. 5. Classified dye patterns on horizontal cross-sections for the soil monoliths (a) Merzenhausen #1 and (b) Merzenhausen #2. The black areas are most intensively stained and white is non-stained. The symbols indicate the selective soil sampling locations.

mass exchange leading to divergence of the transport pathways. The first 5 cm of the MRZ monoliths can be considered as the attractor zone, where a redistribution of BB into preferential transport pathways was initiated. This coincided with that part of the soil that was loosened. Below, the transmission zone started where BB transport occurred in more or less isolated flow regions. These stained flow regions correlated well with the pattern of the higher intensity classes in the attractor zone and persisted over depth. The limited duration of BB application prohibited the development of transport pathways deeper down the profile in MRZ#1. Due to the limited length of the soil monoliths, the dispersion zone is absent.

The dye patterns were generally smooth in the MRZ monoliths which hinted at a rather large spatial correlation length for the dye concentration compared to the diameter of the monolith, especially with respect to the patterns observed in the SCH monolith (Fig. 6). Kasteel et al. (2005) performed a field-plot dye tracer infiltration

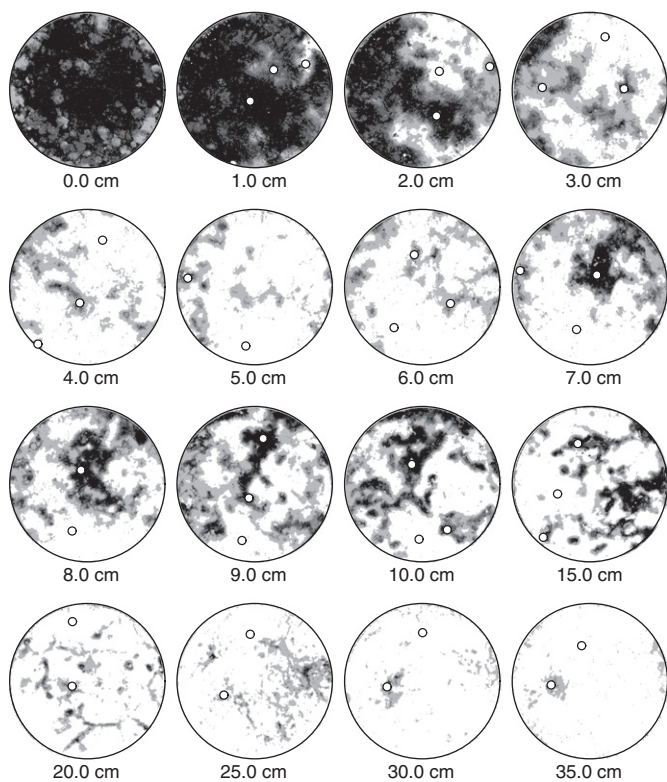


Fig. 6. Classified dye patterns on horizontal cross-sections for the soil monolith Scheyern. The black areas are most intensively stained and white is non-stained. The symbols indicate the selective soil sampling locations.

experiment in the same MRZ soil. For the upper 30 cm of the soil profile, they found a correlation length for the dye concentration of 5 to 7 cm on several $1 \times 1 \text{ m}^2$ plots. No such data were available for the SCH soil, but this soil is more macroporous and the observed dye patterns are a reflection of this, resulting in an obviously much smaller spatial correlation length for the dye concentration. The dye patterns in both soils indirectly show that physical non-equilibrium processes were likely to be more dominant in the SCH monolith, thereby affecting transport processes in a more pronounced way. Rapid local velocities induced by physical non-equilibrium may reduce sorption if it is kinetically limited (thereby promoting chemical non-equilibrium). This is reflected in a better prediction of the mean travel distance of benazolin using the K_{oc} -concept compared to BaP.

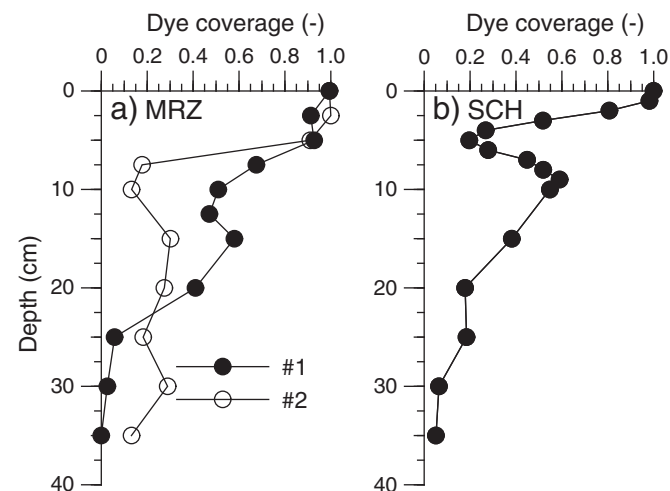


Fig. 7. Depth profiles of dye coverage for (a) the Merzenhausen (MRZ) soil monoliths and (b) the Scheyern (SCH) monolith.

Dye coverage was also nearly 100% close to the soil surface for the SCH monolith (Fig. 7b). The homogenization process by loosening the upper 5 cm was less effective for loam in the SCH monolith compared to the silty loam in the MRZ monolith, resulting in an only 1 to 2 cm thick attractor zone, with more small-scale structures in the concentration patterns. A less deep complete staining of the upper part of the SCH monolith can also be attributed to the supposedly larger BB sorption capacity, which is strongly positively correlated with clay content (Ketelsen and Meyer-Windel, 1999; Morris and Mooney, 2008).

The transmission zone started below 1 to 2 cm depth, but was structured differently compared to the MRZ monoliths. A remarkable decrease in dye coverage to only 20 to 30% was observed between 4 and 6 cm depth. A few black spots (highest intensity class) marked macropores, the majority of the stained pixels belonged to the less intense dye intensity classes. Dye coverage then increased to almost 60% at 10 cm depth, due to a redistribution of dye initialized at 7 cm depth by diverging transport pathways. Below this redistribution zone, a large part of the stained pixels belonged to the highest intensity class. This is an indication that these pathways were fed directly by the input solution without much dilution by mixing processes. Dye coverage rapidly decreased below 20 cm depth. An important feature was a single continuous root channel, which was stained blue. This indicated that this root channel conducted water and was most likely responsible for the rapid increase of BB and its high concentration in the drainage water (Fig. 4b).

The percentage of the bulk soil volume in the monolith that is stained with BB can be deduced from the dye coverage using the trapezoidal rule. Note that the dye stained soil volume is not identical to the transport volume, which is the soil volume that participates to solute transport, since the volume of the soil matrix is excluded in the definition of the latter. From these calculations it follows that the dye tracer stained more volume in the MRZ monoliths (42% for MRZ#1 and 36% for MRZ#2) than in the SCH monolith (31%). This is in line with the slightly larger averaged volumetric water content and the supposedly larger sorption in the SCH monolith.

3.5. Selective sampling

For each horizontal cross-section, we selectively took one sample of a few grams from the non-stained areas and one to three samples from the stained areas (see Fig. 5 and Fig. 6 for the exact sampling locations). We analyzed the samples for total ^{14}C -activity and the results are shown in Fig. 8a and c for benazolin (MRZ) and in Fig. 9a for BaP (SCH), together with the bulk soil total ^{14}C -activity in the slices.

For heterogeneous concentration distribution in soils, it is obvious that the analysis of total ^{14}C -activity in larger soil volumes (slices) diluted the concentration, even below the detection limit (see between 20 and 30 cm depth for benazolin in Fig. 8c and below 25 cm depth for BaP in Fig. 9a), because contaminated soil is mixed with uncontaminated soil. Thus, local concentrations might be much higher than expected from concentration analyses in bulk soil. These differences were more pronounced deeper in the monoliths where smaller parts of the horizontal cross-sections were stained. For all depths and all concentrations, the lowest measured local concentration could always be assigned to the samples chosen from the non-stained areas.

For the mobile leacher benazolin in the MRZ monolith, we found total ^{14}C -activity in each selectively sampled soil sample, except for the deepest sampling location in MRZ#1, irrespective whether these sampling locations were stained or non-stained. For the nonleacher BaP in the SCH monolith, BB was a much better indicator for possible detection of ^{14}C -activity deeper than 15 cm below the soil surface, with only a few water-conducting preferential transport pathways. Here, total ^{14}C -activity was consistently below the limit of detection in the non-stained areas. The strong sorption of BaP limits its lateral mass exchange, which is accentuated when its transport occurs mainly in isolated transport pathways.

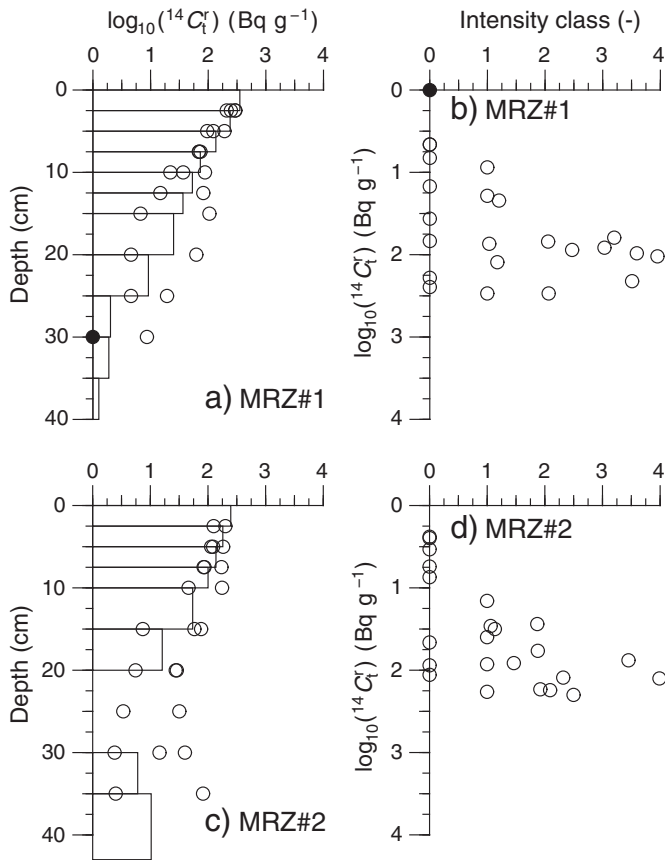


Fig. 8. Depth profiles of the bulk soil ^{14}C -benazolin concentration in the soil slices (bars) and local ^{14}C -concentration in the selective sampling spots (symbols) (a, c) and the local ^{14}C -benazolin concentration vs. Brilliant Blue intensity class (b, d) in the two Merzenhausen (MRZ) soil monoliths. Filled symbols refer to samples where the ^{14}C -concentration is below the detection limit of 0.8 Bq g^{-1} .

A stained root channel was visible between the 20-cm depth and the bottom of the SCH monolith (Fig. 6). We selectively sampled this root channel and found rather high concentrations of ^{14}C -BaP in the stained soil samples (Fig. 9a). Unfortunately, the amount of sampled soil was too low to analyze whether the parent compound BaP or a more mobile metabolite was transported down to the 25-cm depth in this root channel. However, since we mainly found the parent compound BaP at 10 cm depth, it is very likely that it was BaP itself that

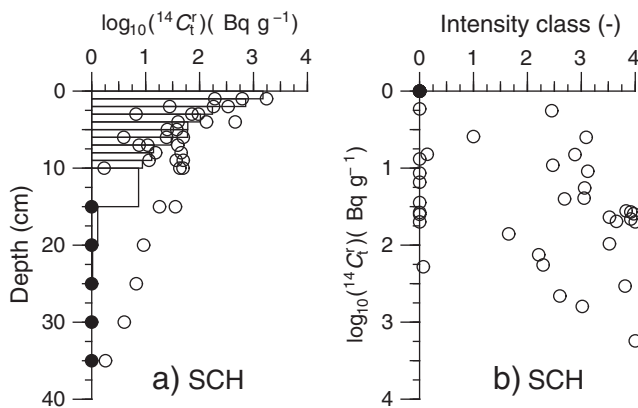


Fig. 9. Depth profiles of the bulk soil ^{14}C -benzo[a]pyrene concentration in the soil slices (bars) and local ^{14}C -concentration in the selective sampling spots (symbols) (a) and the local ^{14}C -benzo[a]pyrene concentration vs. Brilliant Blue intensity class (b) in the Scheyern (SCH) soil monolith. Filled symbols refer to samples where the ^{14}C -concentration is below the detection limit of 0.8 Bq g^{-1} .

was transported to these depths. This exemplifies that root channels, or macropores in general, may act as short-cuts for the transport of virtually immobile compounds, bypassing large volumes of the bulk soil (e.g. Flury, 1996; Kasteel et al., 2010; Nielsen et al., 2011; Vereecken, 2005). However, the leached mass flux density was very low.

The relationship between the ^{14}C -activity and the dye intensity class is shown in Fig. 8b and d for benazolin and in Fig. 9b for BaP. Notice that both compounds were applied earlier compared to BB, so that the characteristic time scale of lateral mass transfer was longer than for BB. Therefore, the local-scale ^{14}C -activity and intensity class were poorly correlated for all monoliths. Nevertheless, the spread of benazolin over all intensity classes accentuated the larger lateral mass transfer for the more mobile herbicide benazolin, which had more time to reach less accessible flow regions than BB. Even the nonleacher BaP reached areas where no BB was assigned to in the upper part of the soil monolith. Notice that this does not necessarily mean that no BB was transported to these locations, because its detection limit with the image analysis method is certainly much higher than for the ^{14}C -technique.

4. Conclusions

Visualization of transport pathways with the food dye Brilliant Blue is a helpful tool for the interpretation of transport behavior of organic compounds with different sorption properties. The staining pattern can identify preferred transport pathways or hydrologically active preferential transport pathways, which can also be used to guide a selective sampling scheme in case of less mobile compounds. Fast breakthrough of all compounds in the 40-cm long monoliths after only 1 to 2 cm of cumulative infiltration was due to by-passing larger parts of the soil matrix. The transport of the nonleacher BaP cannot be explained by its chemical properties only. Structure-dependent flow, in this case through a root channel, is the key process determining the movement of the virtually immobile BaP, although mass flux density was low. Physical non-equilibrium processes – that additionally may reduce sorption if it is kinetically limited and thereby promoting chemical non-equilibrium processes – are likely to be a major factor for transport in structured soils. Therefore, soil structure cannot be neglected to assess the fate of nonleachers in undisturbed soils.

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